Jonathan Tybirk

Copenhagen, Denmark

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Education

- **Technical University of Denmark (DTU)** *Sep 2023 – Jun 2027 (expected)*
  Bachelor of Science in Artificial Intelligence and Data GPA: 10.6 / 12
  - Summer exchange, Nanyang Technological University, Singapore (2025)

Research Experience

- **Flow-Matching World Model for 3D Perception** *Jun 2026 – Aug 2026*
  Undergraduate Researcher, Hong Kong University of Science and Technology Hong Kong
  - Developed a generative world model that predicts how a partially observed 3D scene will appear from a new viewpoint, enabling a robot to anticipate what it would perceive after moving
  - Built the research pipeline from simulated multi-view LiDAR data through model training and rollout evaluation; trained experiments on a high-performance computing cluster
- **Equivariant Learning for 3D Center-of-Mass Prediction** *Feb 2025 – Jun 2025*
  Research Project in Geometric Deep Learning, Technical University of Denmark Lyngby, Denmark
  - Implemented SE(3)-equivariant and conventional graph neural networks in PyTorch and e3nn to predict object centers of mass from sparse LiDAR scans
  - Built a reproducible ShapeNet pipeline for synthetic LiDAR generation, training, and evaluation; showed that rotational data augmentation matched predictive accuracy, while architectural equivariance guaranteed exact geometric consistency

Experience

- **HIVE Robots** *Jan 2026 – Present*
  Robotics Machine Learning Engineer Copenhagen, Denmark
  - Developed teleoperated and autonomous demonstrations on Unitree G1 humanoid robots for navigation, component sorting, and tool-assisted disassembly; autonomous policies outperformed expert teleoperation on some tasks
  - Built and operated the learning pipeline from teleoperation data capture through VLA post-training and autonomous evaluation, including fine-tuning NVIDIA GR00T for real-world manipulation
- **DTU RAVEN — Autonomous Drone Team** *Dec 2025 – Present*
  Autonomy Software Team Member Copenhagen, Denmark
  - Developed autonomous area-coverage and target-localization missions for a custom-built quadcopter, including route optimization, safety constraints, and payload-drop integration
  - Won the 2026 C-UASC autonomous drone competition in California, setting the competition's all-time speed record and achieving the highest waypoint accuracy
- **Technical University of Denmark** *Jan 2024 – May 2026*
  Teaching Assistant, Computer Science and Artificial Intelligence Lyngby, Denmark
  - Taught programming, statistics, machine learning, and symbolic AI courses; supervised first-year projects in deep reinforcement learning, convolutional neural networks, and text embeddings

Selected Awards

- **Case Winner, Audio Explorers Engineering Competition, Demant, 2026:** won the multi-talker speech-separation software case with a solo submission; completed the winners' study programme in Toronto
- **Danish National Student Championship in Artificial Intelligence, 2025:** winning team; developed the winning healthcare subchallenge solution for evaluating medical statements

Technical Skills

Programming: Python, C++, SQL
Machine learning: PyTorch, transformer models, VLA models, diffusion and flow matching, geometric deep learning
Robotics and systems: ROS 2, Linux, Git, Docker, high-performance computing

Leadership

- **Danish Student Association for Rocketry** *Sep 2024 – Apr 2026*
  Treasurer and Board Member Lyngby, Denmark